

Zixuan Guo

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Biography

Audio AI and Music Technology researcher-developer with a background in Computer Science and experience building machine learning systems for music, audio, and interactive media. Research interests include controllable music generation, robust machine listening, audio representation learning, and human-centered AI tools for creative workflows. Currently pursuing an M.M. in Music Technology at NYU.

EDUCATION

New York University

New York, NY

Master of Music (M.M.) in Music Technology

Expected May 2026

GPA: 4.0/4.0 | **Coursework:** Deep Learning for Media, Machine Listening, Digital Signal Processing, Music Information Retrieval, C++ Audio Application Development

Jilin University

Jilin, China

Bachelor of Science in Computer Science and Technology

Aug 2019 – Jun 2024

GPA: 3.41/4.0 | **Coursework:** Object-Oriented Programming, Software Engineering, Algorithms & Data Structures, Operating Systems, Machine Learning, Linux Systems

PREPRINTS

1. Guo, Z., Zhong, J., & Chen, S. (2026). Noise-robust instrument classification under real-world SNR conditions. Preprint submitted to SSRN.

AWARDS

1. **Academic Achievement Recognition**, Graduate Music Technology Program, NYU Steinhardt *May 2026*
2. **Third-Class Scholarship**, Department of Computer Science and Technology, Jilin University *Sep 2020*

RESEARCH EXPERIENCE

Robust Acoustic Scene Classification Research (DCASE / ConvMixer-RSC)

Spring 2026

- Investigated robustness and domain generalization in acoustic scene classification using ConvMixer and Representation Self-Challenging (RSC) under reduced-data and cross-device conditions.
- Built a reproducible DCASE-style benchmarking pipeline with deterministic training, experiment tracking, checkpointing, and per-class evaluation.
- Analyzed ConvMixer token/channel mixing behavior and explored Efficient Channel Attention (ECA) integration for improved inter-channel representation learning.
- Developed modular audio frontend interfaces for learnable representations (e.g., LEAF) and domain-robust preprocessing.
- GitHub Link

Audio Backend Developer | XRcising (Mixed-Reality Fitness Platform)

Spring 2026

- Proposed an audio feature extraction and event generation pipeline converting user-uploaded music into time-aligned XR interaction targets for Meta Quest.

- Designed a prototype MIR pipeline integrating beat tracking, spectral features, and source separation (nussl) for spatial interaction mapping.

Generative AI Developer | Master's Thesis: Controllable Symbolic Music Generation

May 2026

- Designed and implemented a DAW-integrated controllable symbolic music generation system combining probabilistic generation with deterministic symbolic control.
- Developed modular generation workflows for multi-track continuation and transformation using Transformer-based and latent-space models.
- Built a JUCE-based plugin and Python inference pipeline enabling interactive AI-assisted music generation inside DAW workflows.
- Evaluated controllability and generation consistency through symbolic post-processing, latent-space manipulation, and iterative system-level testing.
- [GitHub Link](#)

Pitch Tracking in Non-Ideal Conditions

Fall 2025

- Developed a benchmarking framework for evaluating pitch tracking algorithms under noise, distortion, and tuning variations.
- Built audio augmentation pipelines and standardized evaluation workflows across CREPE, Basic Pitch, and Librosa.
- Designed a unified evaluation system using MIR metrics including OA, RPA, RCA, and VR.
- [GitHub Link](#)

Software Developer | ML Pipeline Modernization (DeepSqueak)

Summer 2025

- Migrated acoustic event detection from YOLOv2 to anchor-free YOLOX for spectrogram-based audio inputs.
- Developed MATLAB preprocessing pipelines for multi-channel spectrogram generation, noise augmentation, and hard-negative mining.
- Improved low-SNR robustness through targeted augmentation and training-data refinement.
- [GitHub Link](#)

Noise-Robust Musical Instrument Classification

Spring 2025

- Developed a noise-robust musical instrument classification system for low-SNR environments.
- Designed a curriculum-learning pipeline with progressive SNR degradation and CLAP-based feature representations.
- Implemented TCN- and Transformer-based architectures with harmonic-enhanced fallback mechanisms for extreme-noise robustness.
- [GitHub Link](#)

Machine Learning Developer | Bachelor's Thesis: Domain-Adversarial Transfer Learning

Spring 2024

- Architected a cross-domain transfer learning pipeline using Domain-Adversarial Neural Networks (DANN).
- Developed intermediate-domain generation modules via Mixup and CutMixup augmentation for improved domain generalization.
- Evaluated the system on Office-31 and OfficeHome datasets, outperforming traditional transfer-learning baselines.
- [GitHub Link](#)

MUSIC TECHNOLOGY WORKS COLLECTIONS

Audio Visualizer JUCE Plugin

December 2025

- Audio Visualizer is a real-time audiovisual plugin built with JUCE, translating live audio into an expressive, physically inspired particle system inside a DAW. The plugin runs entirely in real time, analyzing incoming audio on the audio thread and mapping continuous control signals to a custom visual engine that reacts to musical energy and timbral change. Its particle-orbit visual model drives motion, scale, color, and rotation through smoothed audio parameters. An optional Emotion mode uses learned latent controls to shape higher-level visual behaviors, allowing the visualization to feel responsive rather than purely reactive.
- [Github Link](#)

Monophonic MIDI Synthesizer (Max/MSP)

December 2025

- An interactive and visualized monophonic MIDI synthesizer built in Max/MSP, designed as a playable instrument that supports real-time performance through both a virtual on-screen keyboard and external MIDI controllers. The synthesizer integrates waveform selection, ADSR envelope shaping, LFO-based modulation (frequency and depth), nonlinear overdrive, filtering, and time-based effects, with each processing module individually controllable and bypassable.
- [Github Link](#)

Interactive Music Generation via Image-Based Audio Parameter Mapping

September 2024

- This project explores real-time music generation driven by image features, using Max/MSP to convert visual information (color, brightness, saturation, and spatial position) into dynamic audio parameters. By continuously analyzing the color block at the user's current mouse position, the system creates an interactive audiovisual environment where "what you see" and where you choose to look directly shapes "what you hear."
- [Google Drive Link](#)

MUSIC WORKS COLLECTIONS

Misty Forest

December 2024

- Live sound processing performance using Max/MSP, integrating pre-recorded clarinet material with real-time MIDI-controlled transformations.
- Designed multi-patch system combining spectral processing (FFT), delay, filtering, and algorithmic control to create an interactive and immersive sonic texture.
- Structured the performance into multiple controllable segments with a text-based score, enabling real-time interaction and adaptive sound evolution during live execution.
- [Google Drive Link](#)

Walking Through the Dark

November 2024

- Algorithmic composition developed in SuperCollider, constructing multiple synthesized sound modules including pads, arpeggios, and noise-based textures.
- Designed a modular synthesis system combining oscillators, frequency modulation, filtering, and envelope control to create evolving and narrative-driven soundscapes.
- [Google Drive Link](#)

Playing Bass in a Mountain

February 2024

- Mixed-media composition combining live bass performance with sound transformation and electronic production in Logic Pro.
- Designed a sound design pipeline transforming bass guitar techniques into pads, percussive elements, and synthesized textures using delay, reverb, and granular effects.

- Structured the piece into multiple narrative-driven sections with automated parameter control, creating an evolving and immersive sonic environment.
- [Google Drive Link](#)

Breaking Free

January 2024

- Composed, produced, and performed an original EDM track, integrating MIDI arrangement, live guitar recording, and vocal production across Logic Pro and Ableton Live.
- Designed full production workflow including sound selection, arrangement, vocal recording, and mixing/mastering for a genre-oriented electronic music piece.
- Developed dynamic song structure combining verse, build-up, and drop sections, incorporating layered synthesis and instrumental design to create contrast and energy progression.
- [Google Drive Link](#)

The Evil Within 2 Trailer Soundtrack

October 2023

- Composed and produced a full trailer soundtrack for a psychological horror game, designing a multi-section structure (intro, bridge, climax, ending) synchronized to visual narrative.
- Integrated sound design and music composition, combining synthesized textures, vocal recording, and environmental sound effects to create immersive horror atmospheres.
- Applied advanced audio processing techniques including EQ automation, spatial layering, and dynamic arrangement to enhance tension and cinematic impact.
- [Google Drive Link](#)

Dancing in the Rain

April 2023

- Composed an original solo piano piece inspired by rainy-day scenes in Changsha, focusing on emotional expression and atmospheric storytelling.
- Explored melodic phrasing and harmonic progression to evoke a sense of solitude and introspection.
- [NetEase Cloud Music Link](#)

Lonely Travel

September 2022

- Composed and produced an experimental electronic piece in Ableton Live, inspired by real-world narrative scenarios and structured into multi-stage storytelling sections.
- Designed custom synthesizer patches using LFO-controlled modulation and filtering to generate evolving ambient textures and stochastic timbral variations.
- Transformed environmental audio samples into musical elements through audio effects processing (resonator, flanger, chorus), integrating sound design with compositional structure.
- [SoundCloud Link](#)

TECHNICAL SKILLS

1. *Programming*: Python, C/C++ (JUICE), MATLAB, SQL
2. *Machine Learning*: PyTorch, Transformers, Diffusion Models, VAEs, scikit-learn, Representation Learning
3. *Audio & MIR*: Music Information Retrieval, Audio DSP, Machine Listening, Source Separation, Symbolic Music Generation
4. *Systems & Tools*: Linux/Unix, Git, Docker, REST APIs, Experiment Reproducibility